

TechPratham BigBlueButton (BBB) Failed Recording Publish & Republish Runbook

Purpose: A repeatable, production-safe procedure for diagnosing and republishing BBB recordings when presentation publishing fails. This documents the incident resolved on 8 September 2026 and the exact recovery method used.

1. Incident Summary

28 recordings had failed at the **presentation publish** stage. Earlier recording-processing stages had completed, but the presentation publisher could not create/use the expected **logo.png**. It then failed while calculating presentation image dimensions.

Root-cause chain: missing logo.png → ImageMagick **xc:transparent** generation failed → no valid logo dimensions → Ruby publisher raised *nil can't be coerced into Float* in **panzoom_viewbox** → **metadata.xml** was not produced.

2. Production-Safety Rules

- Do not reboot during live classes.
- Do not restart BBB, FreeSWITCH, Redis, Nginx, or other services just to republish an old recording.
- Process recordings one at a time.
- Do not delete failed staging backups until playback is verified.
- Make backups before changing BBB scripts or ImageMagick configuration.

3. File Structure

```
/var/bigbluebutton/  
■■■ recording/  
■   ■■■ process/presentation/<MEETING_ID>/  
■   ■   ■■■ presentation/logo.png  
■   ■■■ publish/presentation/<MEETING_ID>/      # staging  
■   ■■■ status/published/  
■       ■■■ <MEETING_ID>-presentation.fail  
■       ■■■ <MEETING_ID>-presentation.done  
■■■ published/presentation/<MEETING_ID>/  
    ■■■ metadata.xml  
    ■■■ slides/  
    ■■■ ...  
  
/usr/local/bigbluebutton/core/scripts/  
■■■ publish/presentation.rb  
  
/var/log/bigbluebutton/presentation/  
■■■ publish-<MEETING_ID>.log
```

4. Find Failed Publishes

```
sudo find /var/bigbluebutton/recording/status/published \  
-type f -name "*.fail" -printf '%f\n' | sort
```

Raw count:

```
echo "FAIL markers: $(sudo find /var/bigbluebutton/recording/status/published -type f -name '*.fail' | wc -l)
```

Important: an old .fail can remain after a successful retry, so a raw .fail count is not an unresolved-failure count.

5. Find Only Unresolved Failures

```
for f in $(sudo find /var/bigbluebutton/recording/status/published \  
-type f -name '*-presentation.fail'); do  
  donefile="${f%.fail}.done"  
  [ -f "$donefile" ] || echo "$f"  
done  
  
echo "Unresolved failures: $(for f in $(sudo find /var/bigbluebutton/recording/status/published -type f -name '*-presentation.f
```

6. Check the Processing Directory

```
MEETING="ecde3596e7901f6731d8ba637fd288032cc91535-1788877202224"
ls -lah "/var/bigbluebutton/recording/process/presentation/$MEETING/presentation/"
ls -l "/var/bigbluebutton/recording/process/presentation/$MEETING/presentation/logo.png"
```

During the incident, all 28 failed recordings were missing this logo file.

7. Read the Publisher Log

```
sudo less "/var/log/bigbluebutton/presentation/publish-$MEETING.log"

sudo grep -nE "Missing image|convert-im6|no images defined|nil can't be coerced|panzoom_viewbox|ERROR|Finished publishing" \
"/var/log/bigbluebutton/presentation/publish-$MEETING.log"
```

The incident log showed the missing logo, ImageMagick canvas failure, and the downstream Ruby Float error.

8. Diagnose ImageMagick

```
which convert
convert -version
convert -list configure | grep -iE 'DELEGATES|CODER_PATH|CONFIGURE_PATH'
convert -list format | grep -iE '^ *XC|CANVAS'
```

ImageMagick was installed and its xc module existed, but **convert -size 1600x1200 xc:transparent ...** failed. We did not reinstall packages or make global changes during the live incident.

9. Create a Valid Blank PNG Without ImageMagick

We used Python's standard library to create a transparent 1600×1200 RGBA PNG, avoiding the broken ImageMagick canvas operation.

```
python3 - <<'PY'
import struct, zlib

width, height = 1600, 1200
path = "/tmp/bbb-test-logo.png"

def chunk(name, data):
    return (struct.pack(">I", len(data)) + name + data +
            struct.pack(">I", zlib.crc32(name + data) & 0xffffffff))

raw = b"".join(
    b"\x00" + (b"\x00" * (width * 4))
    for _ in range(height)
)

png = (b"\x89PNG\r\n\x1a\n"
      + chunk(b"IHDR", struct.pack(">IIBBBBB", width, height, 8, 6, 0, 0, 0))
      + chunk(b>IDAT", zlib.compress(raw, 9))
      + chunk(b"IEND", b""))

with open(path, "wb") as f:
    f.write(png)

print("Created:", path)
print("Size:", len(png), "bytes")
PY

file /tmp/bbb-test-logo.png
identify /tmp/bbb-test-logo.png
```

Expected: PNG image data, **1600 × 1200**, 8-bit/color RGBA.

10. Install the Blank Logo Into the Failed Recording

```
sudo mkdir -p \
"/var/bigbluebutton/recording/process/presentation/$MEETING/presentation"

sudo cp /tmp/bbb-test-logo.png \
"/var/bigbluebutton/recording/process/presentation/$MEETING/presentation/logo.png"

sudo chown bigbluebutton:bigbluebutton \
"/var/bigbluebutton/recording/process/presentation/$MEETING/presentation/logo.png"

sudo chmod 644 \
"/var/bigbluebutton/recording/process/presentation/$MEETING/presentation/logo.png"

sudo -u bigbluebutton identify \
```

```
"/var/bigbluebutton/recording/process/presentation/$MEETING/presentation/logo.png"
```

11. Back Up Existing Publish Staging

```
sudo mv \
"/var/bigbluebutton/recording/publish/presentation/$MEETING" \
"/var/bigbluebutton/recording/publish/presentation/${MEETING}.failed-backup-$(date +%Y%m%d-%H%M%S)"
```

This preserves the failed staging data and makes the retry reversible.

12. Republish Only the Presentation

```
cd /usr/local/bigbluebutton/core/scripts

sudo -u bigbluebutton bundle exec ruby \
  publish/presentation.rb \
  -m "${MEETING}-presentation"
```

This retries only the presentation publisher; it does not require restarting the BBB stack.

13. Verify Publication

```
ls -lah "/var/bigbluebutton/published/presentation/$MEETING/"

test -f "/var/bigbluebutton/published/presentation/$MEETING/metadata.xml" \
  && echo "SUCCESS: metadata.xml exists" \
  || echo "ERROR: metadata.xml missing"

ls -l "/var/bigbluebutton/recording/status/published/${MEETING}-presentation."*
```

A successful retry creates **-presentation.done**. An old .fail may remain as historical evidence.

14. Verify Playback

Open the recording in the BBB/Greenlight interface and confirm that it loads and plays. Keep the timestamped backup until playback has been verified.

15. Batch Repair Script Used

For the 28-recording incident, the recovery script was saved as **/root/repair-bbb-publishes.sh**. It processed one recording at a time, created a logo only when missing, validated it, backed up staging, ran the publisher, checked metadata.xml, and stopped on the first error.

```
chmod +x /root/repair-bbb-publishes.sh
/root/repair-bbb-publishes.sh
```

Result of the incident: **28/28 recordings successfully republished**; the script finished with **ALL SELECTED RECORDINGS PROCESSED**.

16. Safer Future Batch Selection

Do not rerun the old script blindly if .fail files remain. Select only IDs that have .fail without a matching .done:

```
find /var/bigbluebutton/recording/status/published \
  -type f -name '*-presentation.fail' -printf '%f\n' |
sed 's/-presentation\.fail$//' |
while read -r id; do
  if [ ! -f "/var/bigbluebutton/recording/status/published/${id}-presentation.done" ]; then
    echo "$id"
  fi
done
```

17. Failure → Action Reference

Error / Symptom	Action
Missing image file .../logo.png	Check the process presentation directory; if missing, create a valid blank PNG and retry.
convert-im6 ... xc:transparent failed	Treat as an ImageMagick canvas problem; avoid production package changes during live classes.
no images defined	Validate the replacement PNG using file/identify.
nil can't be coerced into Float	Usually a downstream symptom; inspect earlier log lines for the image failure.
metadata.xml missing	Publisher did not complete; inspect log and retry only after fixing the cause.

Error / Symptom	Action
.fail + .done both exist	May be old fail + successful retry. Check .done, metadata.xml, and playback.

18. What NOT to Do During Live Classes

- Do not use bbb-record --rebuildall as the first response to one publish failure.
- Do not restart all BBB services for an old recording failure.
- Do not delete process/publish data before preserving evidence.
- Do not delete .failed-backup-* until playback is verified.
- Do not reinstall ImageMagick or patch BBB publisher code without backup/testing and a maintenance window.

19. Permanent Root-Cause Investigation

The workaround repairs existing recordings. It does not by itself prove that future recordings are safe. The underlying ImageMagick **xc:** operation should be diagnosed separately during maintenance.

```
convert -version
convert -list configure | grep -iE 'DELEGATES|CODER_PATH|CONFIGURE_PATH'
convert -list format | grep -iE '^ *XC|CANVAS'
ldd /usr/lib/x86_64-linux-gnu/ImageMagick-6.9.12/modules-Q16/coders/xc.so
MAGICK_DEBUG=Module convert -size 10x10 xc:transparent null:
```

Only after identifying the cause should the permanent fix be selected: repair ImageMagick, adjust the BBB publisher workflow, or use another supported blank-image method. Test with a new recording before production rollout.

20. Quick Recovery Checklist

- Find *-presentation.fail.
- Identify unresolved failures by checking matching .done.
- Choose one MEETING_ID.
- Read its publish log.
- Check logo.png.
- Create/validate the blank PNG if needed.
- Copy logo.png and set ownership.
- Move staging to a timestamped backup.
- Run presentation.rb only for that recording.
- Confirm metadata.xml.
- Confirm .done.
- Verify playback.
- Then process the next recording.

21. Useful Diagnostics

```
sudo bbb-conf --check
df -h /var/bigbluebutton
df -i /var/bigbluebutton
ls -lh /var/log/bigbluebutton/presentation/
sudo find /var/bigbluebutton/published/presentation \
  -mindepth 1 -maxdepth 1 -type d | wc -l
sudo find /var/bigbluebutton/recording/publish/presentation \
  -maxdepth 1 -type d -name '*.failed-backup-*' -printf '%f\n'
```

22. Incident Record

Date: 8 September 2026

Issue: 28 failed presentation publishes

Observed cause: missing logo.png + ImageMagick xc: failure + downstream nil Float error

Recovery: generated valid transparent PNG, added it to each affected process directory, preserved staging as timestamped backups, reran presentation publisher individually

Result: 28/28 successfully republished

BBB services restarted: No